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Scaling Laws for Precision in High-Dimensional Linear Regression

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Motivation

► Low-precision training is now a scaling variable.

Large-scale models are trained with mixed or low precision for weights, activations, gradients, and optimizer states.

► Empirical low-precision scaling laws disagree on how quantization enters:

- ▷ as reduced effective model or data size,
- ▷ or as an explicit additive risk penalty.

► Existing theoretical scaling laws largely assume full precision.

They do not explain how a concrete low-precision training protocol changes the scaling with model size M and data size N .

What mechanism determines how model size, data size, and numerical precision jointly control scaling laws?

Sketched Linear Regression

► **Model.** We observe Gaussian-sketched covariates $(\mathbf{S}\mathbf{x}, y)$, where $\mathbf{x} \in \mathbb{H} \subseteq \mathbb{R}^p$ and $\mathbf{S}: \mathbb{H} \rightarrow \mathbb{R}^M$ is the Gaussian sketch map and M is the trainable model size.► **Predictor and population risk.** For $\mathbf{v} \in \mathbb{R}^M$,

$$f_{\mathbf{v}}(\mathbf{x}) = \langle \mathbf{v}, \mathbf{S}\mathbf{x} \rangle, \quad \mathcal{R}_M(\mathbf{v}) = \frac{1}{2} \mathbb{E}[(\langle \mathbf{S}\mathbf{x}, \mathbf{v} \rangle - y)^2].$$

► **One-pass quantized SGD** with iterate averaging:

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \gamma g^{(q)}(\mathbf{f}^{(q)}), \quad \bar{\mathbf{v}}_N = \frac{1}{N} \sum_{t=0}^{N-1} \mathbf{v}_t.$$

where $\mathbf{f}^{(q)} = \mathcal{Q}_f(\mathcal{Q}_s(\mathbf{S})\mathcal{Q}_d(\mathbf{x}_t)), \quad g^{(q)} = \mathcal{Q}_o(\mathcal{Q}_l(y_t) - \mathcal{Q}_a((\mathbf{f}^{(q)})^\top \mathcal{Q}_p(\mathbf{v}_{t-1})))$.

Core perspective. Low-precision training requires **all multiplications** to be performed in low-precision formats.

► **Quantization Models**

Unbiased stochastic rounding:

$$\mathbb{E}[\mathcal{Q}_i(\mathbf{u}) \mid \mathbf{u}] = \mathbf{u}, \quad i \in \{d, s, f, l, p, a, o\}.$$

For a vector $\mathbf{x}$ and a matrix $\mathbf{X}$ with PSD matrix $\mathbf{A}$, define

$$\mathbf{D}_{\mathbf{x}} = \mathbb{E}[(\mathcal{Q}(\mathbf{x}) - \mathbf{x})(\mathcal{Q}(\mathbf{x}) - \mathbf{x})^\top \mid \mathbf{x}], \quad \mathbf{D}_{\mathbf{X}}(\mathbf{A}) = \mathbb{E}[(\mathcal{Q}(\mathbf{X}) - \mathbf{X})\mathbf{A}(\mathcal{Q}(\mathbf{X}) - \mathbf{X})^\top \mid \mathbf{X}].$$

Multiplicative quantization

FP-like, signal-dependent error

$$\epsilon \mathbf{x} \mathbf{x}^\top \preceq \mathbf{D}_{\mathbf{x}} \preceq \bar{\epsilon} \mathbf{x} \mathbf{x}^\top$$

$$\epsilon \mathbf{X} \mathbf{A} \mathbf{X}^\top \preceq \mathbf{D}_{\mathbf{X}}(\mathbf{A}) \preceq \bar{\epsilon} \mathbf{X} \mathbf{A} \mathbf{X}^\top$$

Additive quantization

INT-like, signal-independent error

$$\epsilon \mathbf{I} \preceq \mathbf{D}_{\mathbf{x}} \preceq \bar{\epsilon} \mathbf{I}$$

$$\epsilon \text{tr}(\mathbf{A}) \mathbf{I} \preceq \mathbf{D}_{\mathbf{X}}(\mathbf{A}) \preceq \bar{\epsilon} \text{tr}(\mathbf{A}) \mathbf{I}$$

► **Technical assumptions**

Let $\tilde{\mathbf{x}}^{(q)} = \mathcal{Q}_f(\mathcal{Q}_s(\mathbf{S})\mathcal{Q}_d(\mathbf{x}))$ and $\mathbf{H}_f^{(q)} = \mathbb{E}[\tilde{\mathbf{x}}^{(q)}(\tilde{\mathbf{x}}^{(q)})^\top]$.

► **Fourth moment sandwiches.**

For any PSD $\mathbf{A}$,

$$\mathbf{H}_f^{(q)} \mathbf{A} \mathbf{H}_f^{(q)} + \beta \text{tr}(\mathbf{H}_f^{(q)} \mathbf{A}) \mathbf{H}_f^{(q)} \preceq \mathbb{E}[\tilde{\mathbf{x}}^{(q)}(\tilde{\mathbf{x}}^{(q)})^\top \mathbf{A} \tilde{\mathbf{x}}^{(q)}(\tilde{\mathbf{x}}^{(q)})^\top] \preceq \alpha \text{tr}(\mathbf{H}_f^{(q)} \mathbf{A}) \mathbf{H}_f^{(q)}.$$

► **Quantized noise sandwiches.**

Consider $\mathbf{v}^{(q)*} = \arg \min_{\mathbf{v} \in \mathbb{R}^M} \mathbb{E}[(\langle \tilde{\mathbf{x}}^{(q)}, \mathbf{v} \rangle - \mathcal{Q}_l(y))^2]$.

$$\sigma^2 \mathbf{H}_f^{(q)} \preceq \mathbb{E}[\xi^2 \tilde{\mathbf{x}}^{(q)}(\tilde{\mathbf{x}}^{(q)})^\top] \preceq \bar{\sigma}^2 \mathbf{H}_f^{(q)}, \quad \xi = \mathcal{Q}_l(y) - \langle \mathbf{v}^{(q)*}, \tilde{\mathbf{x}}^{(q)} \rangle.$$

► **Scaling-law regime.**

Assume the well-specified model, i.e., $\mathbb{E}[y|\mathbf{x}] = \mathbf{x}^\top \mathbf{w}^*$ and

$\sigma^2 := \mathbb{E}[(y - \mathbf{x}^\top \mathbf{w}^*)^2]$, and the parameter prior, i.e., $\mathbb{E}[\mathbf{w}^* \mathbf{w}^{*\top}] = \mathbf{I}$.

Assume the polynomial data spectrum, i.e., there exists $a > 1$ such that the eigenvalues of $\mathbf{H}$ satisfy $\lambda_i \approx i^{-a}$, $i > 0$.

Scaling Law Upper Bounds

Theorem 1: With probability at least $1 - e^{-\Omega(M)}$ over the randomness of $\mathbf{S}$,

$$\mathcal{R}_M(\bar{\mathbf{v}}_N) \lesssim \frac{1}{M_{\text{eff}}^{a-1}} + \frac{1}{N_{\text{eff}}^{(a-1)/a}} + \sigma^2 + \bar{\epsilon}_3.$$

► **Multiplicative quantization**

$$M_{\text{eff}} = M, \quad N_{\text{eff}} = N \left[\frac{1 + \bar{\epsilon}_2^{(M)}}{(1 - \bar{\epsilon}_3^{(M)})^{\frac{1}{a}}} \right]^{-\frac{a}{a-1}}, \quad \bar{\epsilon}_3 = \bar{\epsilon}_3^{(M)},$$

where $\bar{\epsilon}_2^{(M)} = \bar{\epsilon}_o + \bar{\epsilon}_p + \bar{\epsilon}_a$, $\bar{\epsilon}_3^{(M)} = \bar{\epsilon}_d + \bar{\epsilon}_f + \bar{\epsilon}_s$.

► **Additive quantization**

$$M_{\text{eff}} = M \left[1 + (1 + \bar{\epsilon}_2^{(A)}) \frac{(\bar{\epsilon}_3^{(A)})^2}{1 - \bar{\epsilon}_3^{(A)}} \right]^{-\frac{1}{a-1}}, \quad N_{\text{eff}} = N \left[\frac{1 + \bar{\epsilon}_2^{(A)}}{(1 - \bar{\epsilon}_3^{(A)})^{1/a}} \right]^{-\frac{a}{a-1}}, \quad \bar{\epsilon}_3 = \bar{\epsilon}_3^{(A)},$$

where $\bar{\epsilon}_2^{(A)} = \bar{\epsilon}_a + \bar{\epsilon}_o + \bar{\epsilon}_p [1 + p\bar{\epsilon}_d + M(\bar{\epsilon}_f + \bar{\epsilon}_s + \bar{\epsilon}_s\bar{\epsilon}_d p)]$, $\bar{\epsilon}_3^{(A)} = \frac{\bar{\epsilon}_f + \bar{\epsilon}_s(1 + \bar{\epsilon}_d p) + \bar{\epsilon}_d \frac{p}{M}}{M^{-a} + \bar{\epsilon}_f + \bar{\epsilon}_s(1 + \bar{\epsilon}_d p) + \bar{\epsilon}_d \frac{p}{M}}$.

Two compound effects control precision scaling:

- $\bar{\epsilon}_2$: noise amplification from parameter, activation, and gradient quantization.
- $\bar{\epsilon}_3$: feature-spectrum distortion from data, sketch, and feature quantization.

► **How quantization changes scaling laws.**

- ▷ **Effective data size** N_{eff} : reduced via $\bar{\epsilon}_2$ and $\bar{\epsilon}_3$.
- ▷ **Effective model size** M_{eff} :
 - preserved under **multiplicative quantization**;
 - reduced by $\bar{\epsilon}_2$ and $\bar{\epsilon}_3$ under **additive quantization**.
- ▷ **Additive error** induced by $\bar{\epsilon}_3$.

► **Underlying mechanism.**

- ▷ **Multiplicative quantization:** scales with signal magnitude and preserves the relative spectral structure.
- ▷ **Additive quantization:** creates a constant spectral floor, overwhelms the intrinsic signal in the spectral tail, flattens tail directions, rendering the tail dimensions useless for learning.

Empirical match.

FP formats induce additive error; **INT** formats reduce model size.

Scaling Law Lower Bounds

Theorem 2: Under low-precision well-specific assumption $\mathbb{E}[\xi \mid \tilde{\mathbf{x}}^{(q)}] = 0$, with probability at least $1 - e^{-\Omega(M)}$ over the randomness of $\mathbf{S}$,

$$\mathcal{R}_M(\bar{\mathbf{v}}_N) \gtrsim \frac{1}{(M_{\text{eff}})^{a-1}} + \frac{1}{(N_{\text{eff}})^{(a-1)/a}} + \sigma^2 + (\epsilon_3)^2 + \frac{\epsilon_3}{N}(1 - \bar{\epsilon}_3).$$

► **Multiplicative quantization**

$$N_{\text{eff}} = N \left[\frac{(1 - \bar{\epsilon}_3^{(M)})(1 + \bar{\epsilon}_2^{(M)})}{(1 - \bar{\epsilon}_3^{(M)})^{1/a}} \right]^{-\frac{a}{a-1}}, \quad M_{\text{eff}} = M, \quad \epsilon_3 = \epsilon_3^{(M)},$$

where $\bar{\epsilon}_2^{(M)} = \bar{\epsilon}_o + \bar{\epsilon}_p + \bar{\epsilon}_a$, $\bar{\epsilon}_3^{(M)} = \bar{\epsilon}_d + \bar{\epsilon}_f + \bar{\epsilon}_s$.

► **Additive quantization**

$$N_{\text{eff}} = N \left[\frac{(1 - \bar{\epsilon}_3^{(A)})(1 + \bar{\epsilon}_2^{(A)})}{[1 - N\gamma \left(\frac{1}{1 - \bar{\epsilon}_3^{(A)}} - 1 \right)]^{1/a}} \right]^{-\frac{a}{a-1}}, \quad M_{\text{eff}} = M, \quad \epsilon_3 = \epsilon_3^{(A)},$$

where $\bar{\epsilon}_2^{(A)} = \bar{\epsilon}_a + \bar{\epsilon}_o + \bar{\epsilon}_p [1 + p\bar{\epsilon}_d + M(\bar{\epsilon}_f + \bar{\epsilon}_s\bar{\epsilon}_d p + \bar{\epsilon}_s)]$, $\bar{\epsilon}_3^{(A)} = \frac{\bar{\epsilon}_f + (1 + \bar{\epsilon}_d p)\bar{\epsilon}_s + \bar{\epsilon}_d \frac{p}{M}}{1 + \bar{\epsilon}_f + (1 + \bar{\epsilon}_d p)\bar{\epsilon}_s + \bar{\epsilon}_d \frac{p}{M}}$.

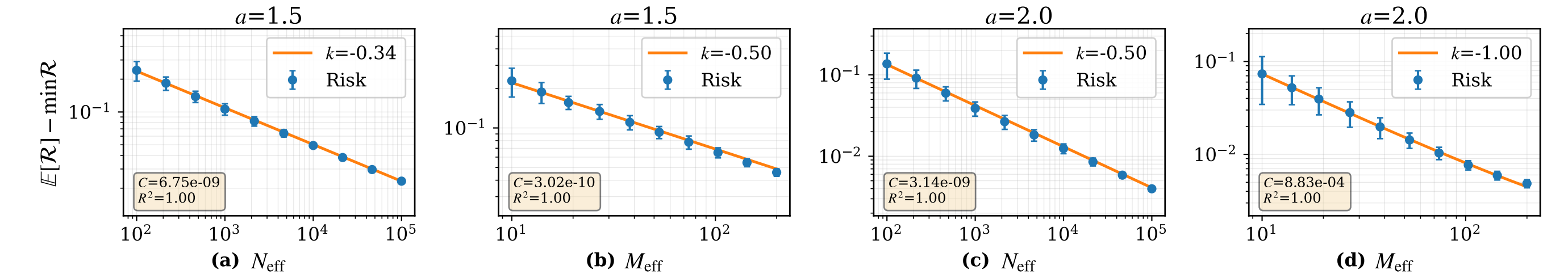
► **What is unavoidable.**

Lower bounds confirm the additive precision floor and the reduction of N_{eff} induced by noise amplification and spectral distortion.

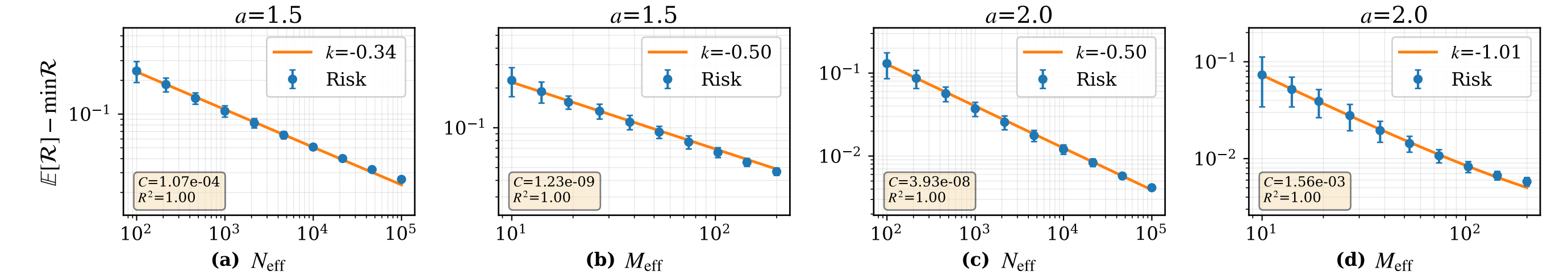
Empirical Validation

Experiment setup.

- Synthetic Gaussian sketched least-squares with polynomial spectrum $\lambda_i \propto i^{-a}$: $(a, p) = (1.5, 10,000)$ and $(2.0, 1,000)$.
- One-pass SGD with iterate averaging, $\gamma = 0.1$, $\sigma = 1$; multiplicative quantization uses $\epsilon = 10^{-3}$ and additive quantization uses $\epsilon = 10^{-8}$.
- Two log-spaced sweeps isolate the scaling variables: fix $M_{\text{eff}} = 2,000$ and vary $N_{\text{eff}} \in [10^2, 10^5]$; fix $N_{\text{eff}} = 20,000$ and vary $M_{\text{eff}} \in [10, 200]$. Each point averages 20 seeds.



Multiplicative quantization. FP-like noise preserves the predicted effective-size scaling.



Additive quantization. INT-like noise follows the same law after replacing M, N by $M_{\text{eff}}, N_{\text{eff}}$.

Results validate the theory.

- Fit excess risk by $\mathcal{R}_M - \frac{1}{2}\sigma^2 = AM_{\text{eff}}^\alpha + BN_{\text{eff}}^\beta + C$.
- The two sweeps separately confirm $\mathcal{R}_M \sim M_{\text{eff}}^{-(a-1)}$ and $\mathcal{R}_M \sim N_{\text{eff}}^{-(a-1)/a}$.

Technical Overview

Challenge: Lower bound analysis for multiplicative quantization.

Multiplicative quantization introduces noise variance proportional to the signal magnitude, creating a **complex feedback loop** where the noise geometry evolves with the iterate $\mathbf{v}_t$. Specifically, the noise-driven covariance $\mathbf{C}_t$ obeys

$$\mathbf{C}_t = \mathbb{E} \left[(1 - \gamma \tilde{\mathbf{x}}_t^{(q)}(\tilde{\mathbf{x}}_t^{(q)})^\top) \mathbf{C}_{t-1} - \gamma \tilde{\mathbf{x}}_t^{(q)}(\tilde{\mathbf{x}}_t^{(q)})^\top \right] + \gamma^2 \Sigma_t,$$

$$\Sigma_t := \mathbb{E} \left[\left(\xi_t + \epsilon_t^{(o)} - \epsilon_t^{(a)} - (\tilde{\mathbf{x}}_t^{(q)})^\top \epsilon_{t-1}^{(p)} \right)^2 \tilde{\mathbf{x}}_t^{(q)}(\tilde{\mathbf{x}}_t^{(q)})^\top \right],$$

where $\epsilon_t^{(p)}, \epsilon_t^{(a)}, \epsilon_t^{(o)}$ are quantization errors defined as the difference of parameter, activation, output gradient and their quantized counterpart, respectively, e.g., $\epsilon_t^{(p)} = \mathcal{Q}_p(\mathbf{v}_t) - \mathbf{v}_t$. Under multiplicative quantization, lower-bounding Σ_t requires iterate-dependent error second moments. For parameter rounding,

$$\mathbb{E}[\epsilon_{t-1}^{(p)}(\epsilon_{t-1}^{(p)})^\top] \succeq \epsilon_p \mathbb{E}[\mathbf{v}_t \mathbf{v}_t^\top].$$

The same issue arises for activation and output-gradient rounding.

A crude lower bound on the iterate.

Drop positive update-noise terms and apply the fourth-moment lower bound:

$$\mathbb{E}[\boldsymbol{\eta}_t \boldsymbol{\eta}_t^\top] \succeq (\mathbf{I} - \gamma \mathbf{H}_f^{(q)}) \mathbb{E}[\boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top] (\mathbf{I} - \gamma \mathbf{H}_f^{(q)}) + \gamma^2 \beta \text{tr}(\mathbf{H}_f^{(q)}) \mathbb{E}[\boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top] \mathbf{H}_f^{(q)}.$$

Unroll the recursion and use the exact mean $\mathbb{E}[\boldsymbol{\eta}_t] = -(\mathbf{I} - \gamma \mathbf{H}_f^{(q)})^t \mathbf{v}^{(q)*}$ to recombine the cross terms:

$$\mathbb{E}[\mathbf{v}_t \mathbf{v}_t^\top] \succeq \frac{\gamma \beta}{2} (\mathbf{I} - \gamma \mathbf{H}_f^{(q)})^{2t} \mathbf{H}_f^{(q)} \|\mathbf{v}^{(q)*}\|_{\mathbf{I} - (\mathbf{I} - \gamma \mathbf{H}_f^{(q)})^{2t}}^2 + (\mathbf{I} - (\mathbf{I} - \gamma \mathbf{H}_f^{(q)})^t) \mathbf{v}^{(q)*} \mathbf{v}^{(q)*\top} (\mathbf{I} - (\mathbf{I} - \gamma \mathbf{H}_f^{(q)})^t).$$

Inserting this time-dependent bound into Σ_t yields the multiplicative-quantization variance lower bound.